Electromagnetism & Electrodynamics HW2

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Technion

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1 Delta function identities

1. show the delta function is symmetric
2. Show
$$\delta\left(\alpha x\right)=\frac{\delta\left(x\right)}{|\alpha|};\quad \alpha\neq 0 \tag{1}$$
3. show
$$\delta\left(g(x)\right)=\sum_{i=1}^{N}\frac{\delta\left(x-x_i\right)}{|g'(x_i)|} \tag{2}$$
4. find
$$\int \mathrm{d}x f(x)\delta'(x) \tag{3}$$
5. show that
$$\delta(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k e^{-ikx} \tag{4}$$

1.1 symmetry

Let $f(x)$ be an arbitrary function. Therefore,

$$\begin{aligned}\int\limits_{-\infty}^{\infty} f(x)\delta(-x)dx = \begin{bmatrix} u=-x \\ du=-dx\end{bmatrix} &= \int\limits_{\infty}^{-\infty} -f(-u)\delta(u)du \\ &= \int\limits_{-\infty}^{\infty} f(-u)\delta(u)du \\ &= f(-0)=f(0)=\int\limits_{-\infty}^{\infty} f(x)\delta(x)dx\end{aligned}$$

Therefore the delta function is symmetric.

1.2 multiplication

$$\begin{aligned}\int\limits_{-\infty}^{\infty} f(x)\delta(\alpha x)dx = \begin{bmatrix} u=\alpha x \\ du=\alpha dx\end{bmatrix} &= \int\limits_{-sgn(\alpha)\cdot\infty}^{sgn(\alpha)\cdot\infty} \frac{1}{\alpha} f\left(\frac{u}{\alpha}\right)\delta(u)du \\ &= \frac{sgn(\alpha)}{\alpha} f\left(\frac{0}{\alpha}\right) \\ &= \frac{1}{|\alpha|} f(0)\end{aligned}$$

Therefore
$$\delta(\alpha x)=\frac{\delta(x)}{|\alpha|}$$

1.3 delta of function

We'll develop $g(x)$ into a taylor series around x_i:

$$g(x) \approx g(x_i) + g'(x_i) \cdot (x - x_i) = g'(x_i) \cdot (x - x_i)$$

ignoring elements beyond first order. Next we'll look at the following integral:

$$\int_{-\infty}^{\infty} \delta(g(x))f(x)dx$$

where $f(x)$ is an arbitrary function. It's clear from the definition of $g(x)$ and the delta function that the integrand is non-zero only at the points x_i, therefore we can integrate only around those points. We'll choose a small enough neighborhood for each point so they are isolated:

$$\int_{-\infty}^{\infty} \delta(g(x))f(x)dx = \sum_{i=1}^{N} \int_{x_i-\varepsilon_i}^{x_i+\varepsilon_i} \delta(g(x))f(x)dx$$

Using the delta function definition and the previous subsection we can find:

$$\begin{aligned}
\sum_{i=1}^{N} \int_{x_i-\varepsilon_i}^{x_i+\varepsilon_i} \delta(g(x))f(x)dx &= \sum_{i=1}^{N} \int_{x_i-\varepsilon_i}^{x_i+\varepsilon_i} \delta(g'(x_i) \cdot (x - x_i))f(x)dx \\
&= \begin{bmatrix} u = g'(x_i) \cdot (x - x_i) \\ du = g'(x_i)dx \end{bmatrix} \\
&= \sum_{i=1}^{N} \int_{sgn(g'(x_i)) \cdot g(x_i-\varepsilon_i)}^{sgn(g'(x_i)) \cdot g(x_i+\varepsilon_i)} \frac{1}{g'(x_i)} \delta(u) f\left(x_i + \frac{u}{g'(x_i)}\right) du \\
&= \sum_{i=1}^{N} \frac{1}{|g'(x_i)|} \int_{g(x_i-\varepsilon_i)}^{g(x_i+\varepsilon_i)} \delta(u) f\left(x_i + \frac{u}{g'(x_i)}\right) du \\
&= \sum_{i=1}^{N} \frac{f(x_i)}{|g'(x_i)|} \\
&= \sum_{i=1}^{N} \int_{-\infty}^{\infty} f(x) \frac{\delta(x - x_i)}{|g'(x_i)|}
\end{aligned}$$

Therefore

$$\delta(g(x)) = \sum_{i=1}^{N} \frac{\delta(x - x_i)}{|g'(x_i)|}$$

1.4 derivative of delta

Integrating by parts we get

$$\int\limits_{-\infty}^{\infty} f(x)\delta'(x)dx = \cancelto{0}{f(x)\delta(x)|_{-\infty}^{\infty}} - \int\limits_{-\infty}^{\infty} f'(x)\delta(x)$$
$$= -f'(0)$$

1.5 fourier transform

If we perform a fourier transform on the delta function we get

$$\mathcal{F}\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x)e^{iky}\mathrm{d}y = \frac{1}{\sqrt{2\pi}} e^{iky}\Big|_{y=0} = \frac{1}{\sqrt{2\pi}} \tag{5}$$

therefore if we apply the inverse fourier transform

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{F}\delta(y)e^{iky}\mathrm{d}y = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iky}\mathrm{d}y \tag{6}$$

and if we rename the parameters

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx}\mathrm{d}k \tag{7}$$

2 Charge Distributions

Write an expression for the volumetric charge distributions for the following distributions in spherical coordinates:

1. A ball of radius R whose center is at the origin and is uniformly charged
2. A spherical shell of radius R placed at the origin, its northern half charged with surface charge σ_N and the southern half charged with σ_S
3. An infinite wire passing through the origin coinciding with $z = y = 0$ and uniformly charged with linear charge density λ

2.1 ball

The volume of a sphere is $V = \frac{4\pi R^3}{3}$ therefore $\rho = \frac{3Q}{4\pi R^3}$ when inside the ball. This means the charge distribution is

$$\rho = \frac{3Q}{4\pi R^3}\Theta(R - r) \tag{8}$$

2.2 shell

A direction calculation

$$\rho = \delta(R - r)\left[\Theta\left(\frac{\pi}{2} - \theta\right)\sigma_N + \Theta\left(\theta - \frac{\pi}{2}\right)\sigma_S\right] \tag{9}$$

2.3 infinite wire

Another direction calculation

$$\rho = \delta\left(\theta - \frac{\pi}{2}\right)[\delta(\phi) + \delta(\phi - \pi)]\lambda(r) \tag{10}$$

$$= \delta\left(\theta - \frac{\pi}{2}\right)[\delta(\phi) + \delta(\phi - \pi)]\frac{\lambda}{r^2} \tag{11}$$

3 Yukawa Potential

The following potential field is given

$$\phi(r) = \frac{1}{r}e^{-Kr}; \quad K > 0 \tag{12}$$

1. Calcualte the gradient for $r \neq 0$
2. Calcualte the laplacian for $r \neq 0$
3. Use gauss' law to find an expression for the laplacian while includes the origin
4. Show that

$$\iiint \mathrm{d}^3x \nabla^2\phi = 0 \tag{13}$$

3.1 gradient

The gradient in spherical coordinates is defined as

$$\boldsymbol{\nabla}\phi = \partial_r\phi\hat{r} + \frac{1}{r}\partial_\theta\phi\hat{\theta} + \frac{1}{r\sin\theta}\partial_\varphi\phi\hat{\varphi}$$

therefore

$$\begin{aligned}\boldsymbol{\nabla}\phi &= \left(-\frac{1}{r^2}e^{-Kr} - \frac{K}{r}e^{-Kr}\right)\hat{r} \\ &= -\left(\frac{1}{r^2} + \frac{K}{r}\right)e^{-Kr}\hat{r}\end{aligned}$$

3.2 laplacian

The laplacian in spherical coordinates is defined as

$$\nabla^2\phi = \frac{1}{r^2}\partial_r(r^2\partial_r\phi) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta\phi) + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2\phi$$

therefore

$$\begin{aligned}\nabla^2\phi &= \frac{1}{r^2}\partial_r(-\left(1+Kr\right)e^{-Kr}) \\ &= \frac{1}{r^2}(-Ke^{-Kr} + \left(1+Kr\right)Ke^{-Kr}) \\ &= \frac{1}{r^2}(K^2re^{-Kr}) \\ &= \frac{K^2}{r}e^{-Kr}\end{aligned}$$

3.3 Gauss' law

Gauss' law for this subsection is

$$\iiint_V \nabla^2\phi d\boldsymbol{x} = \int_{\partial V} \boldsymbol{\nabla}\phi \cdot d\boldsymbol{a}$$

therefore, using the result from subsection 2

$$\iiint_V \nabla^2 \phi d\boldsymbol{x} = \int_{\partial V} -\left(\frac{1}{r^2} + \frac{K}{r}\right) e^{-Kr} \hat{r} \cdot d\boldsymbol{a}$$

We know that $d\boldsymbol{a}$ points in the radial direction therefore $\hat{r} \cdot d\boldsymbol{a} = da$ and thus

$$\begin{aligned}
\iiint_V \nabla^2 \phi d\boldsymbol{x} &= \int_{\partial V} -\left(\frac{1}{r^2} + \frac{K}{r}\right) e^{-Kr} da \\
&= -\left(\frac{1}{R^2} + \frac{K}{R}\right) e^{-KR} \int_{\partial V} da \\
&= -\left(\frac{1}{R^2} + \frac{K}{R}\right) e^{-KR} \cdot 4\pi R^2 \\
&= -4\pi \left(1 + KR\right) e^{-KR}
\end{aligned}$$

Comparing this to a direct calculation

$$\begin{aligned}
\iiint_V \nabla^2 \phi d\boldsymbol{x} &= \iiint_V \frac{K^2}{r} e^{-Kr} d\boldsymbol{x} \\
&= \int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta d\theta \int_0^R r^2 \frac{K^2}{r} e^{-Kr} dr \\
&= 4\pi K^2 \int_0^R r e^{-Kr} dr = \begin{bmatrix} U = r & U' = 1 \\ V' = e^{-Kr} & V = -\frac{1}{K} e^{-Kr} \end{bmatrix} \\
&= 4\pi K^2 \left(-\frac{r e^{-Kr}}{K} \Bigg|_0^R + \frac{1}{K} \int_0^R e^{-Kr} dr \right) \\
&= 4\pi K^2 \left(-\frac{r e^{-Kr}}{K} \Bigg|_0^R + \frac{1}{K} \left(-\frac{1}{K} e^{-Kr} \right) \Bigg|_0^R \right) \\
&= 4\pi K^2 \left(-\frac{R e^{-KR}}{K} + \frac{1}{K} \left(\frac{1}{K} - \frac{1}{K} e^{-KR} \right) \right) \\
&= 4\pi K^2 \left(-\frac{R e^{-KR}}{K} + \frac{1}{K^2} - \frac{1}{K^2} e^{-KR} \right) \\
&= 4\pi \left(-KR e^{-KR} + 1 - e^{-KR} \right) \\
&= 4\pi \left(1 - (1 + KR) e^{-KR} \right)
\end{aligned}$$

We notice that the direct calculation gives an aswer that is 4π off from the result of Gauss' law, which is likely a result of the fact our laplacian definition did not include the origin. Thus we'll add a correction for the origin:

$$\nabla^2 \phi = \frac{K^2}{r} e^{-Kr} - 4\pi \delta^3(\boldsymbol{r})$$

3.4 integral of laplacian

$$\iiint \mathrm{d}^3x \nabla^2 \phi = \iiint \mathrm{d}^3x \left(k^2 \frac{e^{-Kr}}{r} - 4\pi\delta(\mathbf{r}) \right) \tag{14}$$

but

$$\iiint \mathrm{d}^3x k^2 \frac{e^{-Kr}}{r} = 4\pi K^2 \int_0^\infty \mathrm{d}r r e^{-Kr} \tag{15}$$

and since

$$\int_0^\infty \mathrm{d}r r e^{-Kr} = \frac{1}{K^2} \tag{16}$$

then

$$4\pi K^2 \int_0^\infty \mathrm{d}r r e^{-Kr} = 4\pi \tag{17}$$

and the second part

$$\iiint \mathrm{d}^3x 4\pi\delta(\mathbf{r}) = 4\pi \tag{18}$$

adding the two parts together

$$\iiint \mathrm{d}^3x k^2 \frac{e^{-Kr}}{r} = 4\pi K^2 \int_0^\infty \mathrm{d}r r e^{-Kr} = 4\pi - 4\pi = 0 \tag{19}$$